

IP ADDRESSING

- **IP Addressing is Logical Addressing**
- **It works on Network Layer (Layer 3)**
- **Two Versions of Addressing Scheme**
 - **IP version 4 – 32 bit addressing**
 - **IP version 6 – 128 bit addressing**

Bit is a value that will represent 0's or 1's (i.e. Binary)

01010101000001011011111100000001

- **32 bits are divided into 4 Octets known as Dotted Decimal Notation**

First Octet	Second Octet	Third Octet	Fourth Octet
01010101	00000101	10111111	00000001

- **128-bit address is divided along 16-bit boundaries, and each 16-bit block is converted to a 4-digit hexadecimal number and separated by colons (Colon-Hex Notation)**

FEDC:BA98:7654:3210:FEDC:BA98:7654:3210

Taking Example for First Octet :

Total 8 bits, Value will be 0's and 1's

i.e. $2^8 = 256$ combination

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0	0	0	0	0	0	0	0	= 0
0	0	0	0	0	0	0	1	= 1
0	0	0	0	0	0	1	0	= 2
0	0	0	0	0	0	1	1	= 3
0	0	0	0	0	1	0	0	= 4

Total IP Address Range

0 . 0 . 0 . 0

to

255.255.255.255

1 1 1 1 1 1 1 1 = 255

- **Total IP Addressing Scheme is divided into 5 Classes**
 - **CLASS A**
 - **CLASS B**
 - **CLASS C**
 - **CLASS D**
 - **CLASS E**
- LAN & WAN**
- Multicasting**
- Research & Development**
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- To identify the range of each class a bit called priority bit is used.
- Priority Bit is the left most bits in the First Octet
- CLASS A priority bit is **0**
- CLASS B priority bit is **10**
- CLASS C priority bit is **110**
- CLASS D priority bit is **1110**
- CLASS E priority bit is **1111**

For Class A range : First bit of the first octet should be reserved for the priority bit.

0xxxxxxx. xxxxxxxx. xxxxxxxx. xxxxxxxx

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

0 0 0 0 0 0 0 0 = 0

0 0 0 0 0 0 0 1 = 1

0 0 0 0 0 0 1 0 = 2

0 0 0 0 0 0 1 1 = 3

0 0 0 0 0 1 0 0 = 4

0 1 1 1 1 1 1 1 = 127

Class A Range

**0 . 0 . 0 . 0 to
127.255.255.255**

Exception

**0.X.X.X and 127.X.X.X
network are reserved**

For Class B range : First two bits of the first octet should be reserved for the priority bit.

10xxxxxx. xxxxxxxx. xxxxxxxx. xxxxxxxx

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

1 0 0 0 0 0 0 0 = 128

1 0 0 0 0 0 0 1 = 129

1 0 0 0 0 0 1 0 = 130

1 0 0 0 0 0 1 1 = 131

1 0 0 0 0 1 0 0 = 132

1 0 1 1 1 1 1 1 = 191

Class B Range

128. 0 . 0 . 0

to

191.255.255.255

For Class C range : First Three bits of the first octet should be reserved for the priority bit.

110xxxxx. xxxxxxxxxx. xxxxxxxxxx. xxxxxxxxxx

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

1 1 0 0 0 0 0 0 = 192

1 1 0 0 0 0 0 1 = 193

1 1 0 0 0 0 1 0 = 194

1 1 0 0 0 0 1 1 = 195

1 1 0 0 0 1 0 0 = 196

1 1 0 1 1 1 1 1 = 223

Class C Range

192. 0 . 0 . 0

to

223.255.255.255

For Class D range : First four bits of the first octet should be reserved for the priority bit.

1110xxxx. xxxxxxxx. xxxxxxxx. xxxxxxxx

2⁷ 2⁶ 2⁵ 2⁴ 2³ 2² 2¹ 2⁰

1 1 1 0 0 0 0 0 = 224

1 1 1 0 0 0 0 1 = 225

1 1 1 0 0 0 1 0 = 226

1 1 1 0 0 0 1 1 = 227

1 1 1 0 0 1 0 0 = 228

1 1 1 0 1 1 1 1 = 239

Class D Range

224. 0 . 0 . 0

to

239.255.255.255

For Class E range : First four bits of the first octet should be reserved for the priority bit.

1111xxxx. xxxxxxxx. xxxxxxxx. xxxxxxxx

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

1 1 1 1 0 0 0 0 = 240

1 1 1 1 0 0 0 1 = 241

1 1 1 1 0 0 1 0 = 242

1 1 1 1 0 0 1 1 = 243

1 1 1 1 0 1 0 0 = 244

1 1 1 1 1 1 1 1 = 255

Class E Range

240. 0 . 0 . 0

to

255.255.255.255

- **IP address is divided into Network & Host Portion**
- **CLASS A is written as** **N.H.H.H**
- **CLASS B is written as** **N.N.H.H**
- **CLASS C is written as** **N.N.N.H**

- **Class A Octet Format is N.H.H.H**
- **Network bits : 8 Host bits : 24**
- **No. of Networks**
 - = 2^{8-1} (-1 is Priority Bit for Class A)**
 - = 2^7**
 - = $128 - 2$ (-2 is for 0 & 127)**
 - = 126 Networks**
- **No. of Host**
 - = $2^{24} - 2$ (-2 is for Network ID & Broadcast Address)**
 - = $16777216 - 2$**
 - = 16777214 Hosts/Network**

CLASS A
126 Networks
&
16777214 Hosts/Nw

- **Class B Octet Format is N.N.H.H**
- **Network bits : 16 Host bits : 16**
- **No. of Networks**
 - = 2^{16-2} (-2 is Priority Bit for Class B)**
 - = 2^{14}**
 - = 16384 Networks**
- **No. of Host**
 - = $2^{16} - 2$ (-2 is for Network ID)**
 - = 65536 - 2**
 - = 65534 Hosts/Network**

CLASS B
16384 Networks
&
65534 Hosts/Nw

- **Class C Octet Format is N.N.N.H**
- **Network bits : 24 Host bits : 8**
- **No. of Networks**
 - = 2^{24-3} (-3 is Priority Bit for Class C)**
 - = 2^{21}**
 - = 2097152 Networks**
- **No. of Host**
 - = $2^8 - 2$ (-2 is for Network ID)**
 - = 256 - 2**
 - = 254 Hosts/Network**

CLASS C
2097152 Networks
&
254 Hosts/Nw

- **The network address** is represented with all bits as ZERO in the host portion of the address
- **The broadcast address** is represented with all bits as ONES in the host portion of the address
- Valid IP Addresses lie between the Network Address and the Broadcast Address.
- Only Valid IP Addresses are assigned to hosts/clients

Example - Class A

Class A : N.H.H.H

Network Address :

0xxxxxxx.00000000.00000000.00000000

Broadcast Address :

0xxxxxxx.11111111.11111111.11111111

Class A

10.0.0.0

10.0.0.1

10.0.0.2

10.0.0.3

10.255.255.254

10.255.255.255

Network Address

Valid IP Addresses

Broadcast Address

Example - Class B

Class B : N.N.H.H

Network Address :

10xxxxxx.xxxxxxxx.00000000.00000000

Broadcast Address :

10xxxxxx.xxxxxxxx.11111111.11111111

Class B

172.16.0.0

172.16.0.1

172.16.0.2

172.16.0.3

172.16.255.254

172.16.255.255

Network Address

Valid IP Addresses

Broadcast Address

Example - Class C

Class C : N.N.N.H

Network Address :

110xxxxx.xxxxxxxx.xxxxxxxx.00000000

Broadcast Address :

110xxxxx.xxxxxxxx.xxxxxxxx.11111111

Class C

192.168.1.0

192.168.1.1

192.168.1.2

192.168.1.3

192.168.1.254

192.168.1.255

Network Address

Valid IP Addresses

Broadcast Address

- There are certain addresses in each class of IP address that are reserved for LAN. These addresses are called private addresses.
- They can be used for: home & office networks, networks not connected to Internet.

Class A

10.0.0.0 to 10.255.255.255

Class B

172.16.0.0 to 172.31.255.255

Class C

192.168.0.0 to 192.168.255.255

- **Subnet Mask differentiates Network portion and Host Portion**
- **Subnet Mask is been given for host Identification of Network ID**
- **Represented with all 1's in the network portion and with all 0's in the host portion.**

Class A : N.H.H.H

11111111.00000000.00000000.00000000

Default Subnet Mask for Class A is 255.0.0.0

Class B : N.N.H.H

11111111.11111111.00000000.00000000

Default Subnet Mask for Class B is 255.255.0.0

Class C : N.N.N.H

11111111.11111111.11111111.00000000

Default Subnet Mask for Class C is 255.255.255.0